FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014: 2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE JEE (Advanced), 2015

FULL TEST – II

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 246**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section – A (01 – 07)** contains 7 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

   Section – A (08 – 11) contains 4 multiple choice questions which have one or more than one correct answer. Each question carries **+3 marks** for correct answer. There is no negative marking.

   Section-A (12 – 16) contains 2 paragraphs. Based upon paragraph, 2 and 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

2. **Section – B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 marks** will be awarded. There may be one or more than one correct matching. No marks will be given for any wrong matching in any question. There is no negative marking.

3. **Section-C (01 – 06)** contains 6 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+3 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

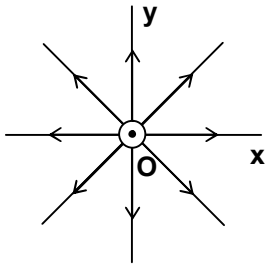
Straight Objective Type

This section contains **7 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** option **is correct**.

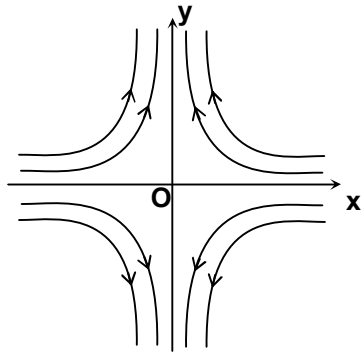
1. Electric field at point P in a region whose coordinates are (x, y, z) is give by : $\vec{E} = E_0 \frac{(-y\hat{i} + x\hat{j})}{x^2 + y^2}$

where E_0 is constant. Which of the following can best represent the Electric field lines corresponding to this field?

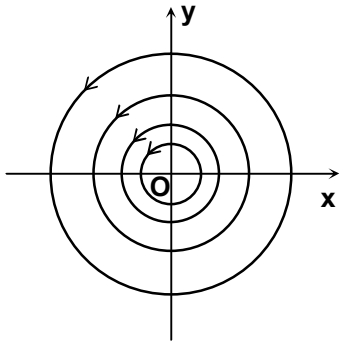
(A)



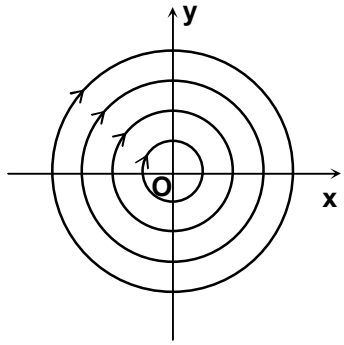
(B)



(C)

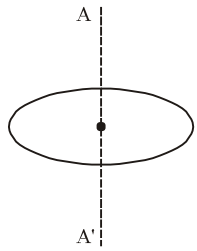


(D)



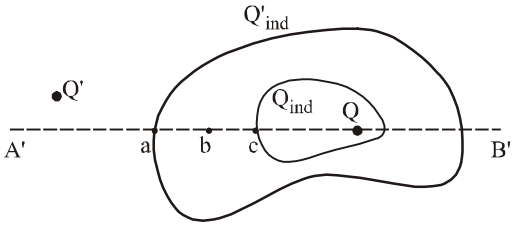
Space for Rough work

2. Net charge distributed on a ring shown in figure is zero. AA' is axis of ring. Then choose the correct statement



(A) Electrostatic field at all points an axis AA' will be zero.
(B) Component of Electrostatic field along axis must be zero.
(C) Component of Electrostatic field perpendicular to the axis must be zero
(D) Electrostatic potential must be zero at all point inside ring in the plane of ring

3. Figure shows a irregular shape conductor with irregular cavity inside it. A charge Q is placed inside cavity and a charge Q' is placed outside conductor. Let Q'_{ind} be charge induced at outside surface of conductor and Q_{ind} be the charge induced at inside surface of cavity. A'B' is an arbitrary line passing through charge Q and a, b, c



be the points on line as shown in figure. Let $\vec{E}, \vec{E}', \vec{E}_{ind}$ and $\vec{E}'_{ind}$ represent electric field at different points due to charge Q, Q' Q_{ind}, & Q'_{ind} respectively. If V_a, V_b and V_c represent potential at point a, b, and c. Choose the incorrect statement

(A) $V_a = V_b = V_c$

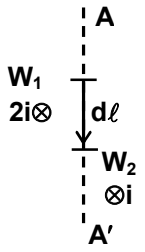
(B) $V_b = \int_{-\infty}^{\vec{r}_b} (\vec{E}+\vec{E}'+\vec{E}_{ind}+\vec{E}'_{ind}).\vec{dr}$, where $\vec{r}_b$ = position vector of point b

(C) $V_b = \int_{-\infty}^{\vec{r}_b} (\vec{E}'+\vec{E}'_{ind}).\vec{dr}$

(D) At point 'b', $\vec{E}' = 0$

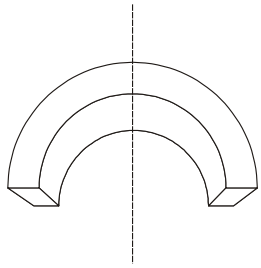
Space for Rough work

4. Two wires W_1 & W_2 carry current '2i' and i respectively directed into the plane of paper as shown in figure. AA' is line at some distance from wire. Let $\vec{B}$ be the net magnetic field due to wires. The magnitude of $\int_{-\infty}^{\infty} \vec{B}.d\vec{\ell}$ will be equal to



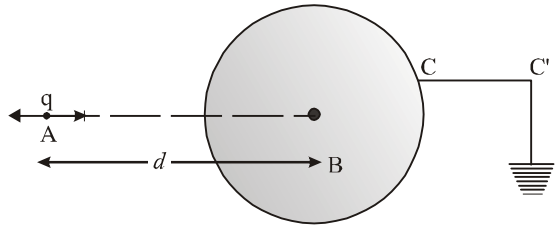
(A) $\mu_0 i/2$ (B) $\mu_0 i/4$
(C) $\mu_0 i/8$ (D) zero

5. A conducting rod of square cross section is bent in a semi circular shape of radius 'r' as shown in figure. Length of side of cross section is equal to 'a'. Resistivity of material is 'ρ'. The resistance of rod across its end will be



(A) $\frac{\pi\rho}{a\ln(1+a/r)}$ (B) $\frac{2\pi\rho}{a\ln(1+a/r)}$
(C) $\frac{\pi\rho(r+a)}{a^2}$ (D) $\frac{2\pi r\rho}{a^2}$

6. A charge 'q' is undergoing S.H.M. about point 'A' along the line AB at distance 'd' from the center B of a conducting sphere of radius R. Amplitude oscillation of charge is 'a' angular frequency is 'ω'. The sphere is grounded through conducting wire CC'. Choose the correct statement



(A) If a < < d then the maximum value of current in wire CC' will be $q\omega\left(\frac{Ra}{d^2}\right)$

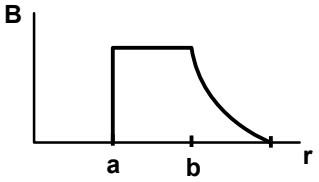
(B) If a < < d then the maximum value of current in wire CC' will be $\left(\frac{q\omega Ra^2}{d^3}\right)$

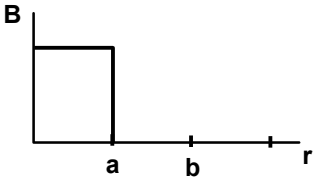
(C) If a < < d then the maximum value of current in wire CC' will be $q\omega\left(\frac{2Ra}{d^2}\right)$

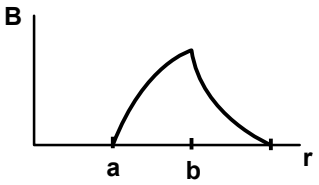
(D) If a < < d then the maximum value of current in wire CC' will be $\left(\frac{2q\omega Ra^2}{d^3}\right)$

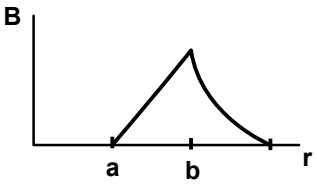
Space for Rough work

7. A hollow cylindrical conductor (inner radius = a, outer radius = b) carries a current i uniformly spread over its cross section. Which graph below correctly gives B as a function of the distance r from the center of the cylinder?

(A) 

(B) 

(C) 

(D) 

(Multiple Correct Choice Type)

This section contains **4 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** are correct.

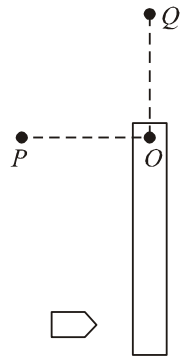
8. A rod is free to rotate in horizontal plane about one of its end hinged at point 'O' as shown in figure. Point P & Q are fixed points located in horizontal plane in such away that PO is perpendicular to rod & QO is along length of rod. A bullet hits the rod perpendicularly at some distance from point 'O' and gets imbedded into it. Then :

(A) Linear momentum of rod + bullet system will decrease after collision.

(B) Linear momentum of rod + bullet system may increase

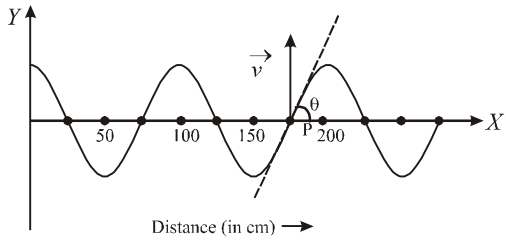
(C) Angular momentum of rod + bullet system about point 'P' will not change before & after collision.

(D) Angular momentum of rod + bullet system about 'Q' may increase after collision.



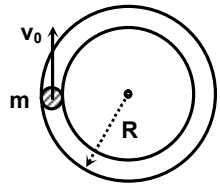
Space for Rough work

9. Shape of string transmitting wave along x-axis at some instant is shown
Velocity of point P is v = 4π cm/s and
$\theta = \tan^{-1}(0.004\pi)$



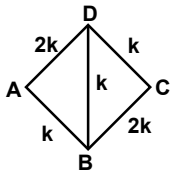
(A) Amplitude of wave is 2 mm
(B) Velocity of wave is 10 m/s
(C) Max acceleration of particle is 80 π^2 cm/sec^2
(D) Wave is traveling in –ve x-direction

10. A small ball of mass m is placed in a circular tube of mass M and radius R in gravity free space. Friction is absent between tube and ball. Ball is given a velocity v_0 as shown. Then,



(A) Path of ball from centre of mass of system will be circular.
(B) Path of ball from centre of mass of system will be elliptical
(C) Radius of curvature of ball at the time of projection of ball is $\frac{MR}{m+M}$
(D) Normal force between tube and ball if M = 2m, at the time of projection of ball is $\frac{2mV_0^2}{3R}$

11. A system of rods is assembled such that each rod has a length ℓ and cross-sectional areas S. The mode of heat transfer is conduction and the system is in steady state. The temperature of Junction A is T and that of C is 2T. Now answer the following question.



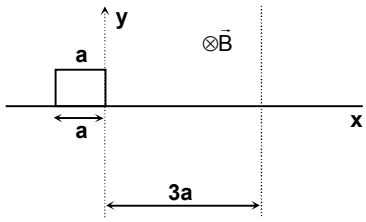
(A) Temperature of junction B is 1.6 T.
(B) Temperature of junction D is 1.4 T.
(C) The rate of heat flow along BD is $\frac{1}{5}\frac{kTS}{\ell}$.
(D) The rate of heat flow along BD is $\frac{21}{5}\frac{kTS}{\ell}$.

Space for Rough work

Comprehension Type

This section contains **2 paragraphs.** Based upon one of the paragraphs **2 multiple choice questions** and based on the other paragraph **3 multiple choice question** have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Paragraph for Questions 12 & 13

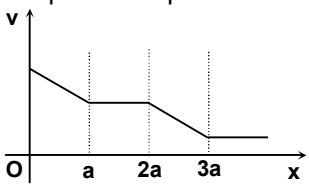


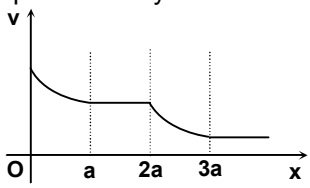
A square wire loop having side a, mass m and resistance R is moving along positive x-axis with speed v_0 enters into uniform steady magnetic field $\vec{B} = B_0(-\hat{k})$ at t = 0 sec as shown in the figure.

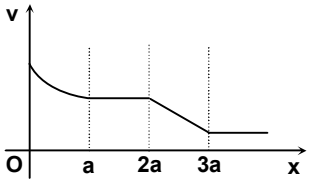
The magnitude of v_0 is sufficient so that the loop comes out from region of magnetic field with v_1. Neglect any type of energy loss other than heat loss in resistances of wires of square (given $v_0 = \frac{3B^2a^3}{mR}$)

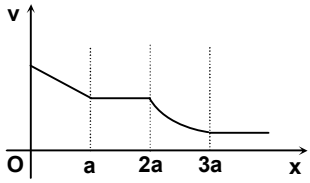
Read above passage carefully and answer the following questions.

12. The plot of speed of square versus position x will be represented by

(A)


(B)


(C)


(D)


13. The total amount of heat loss in the resistance will be

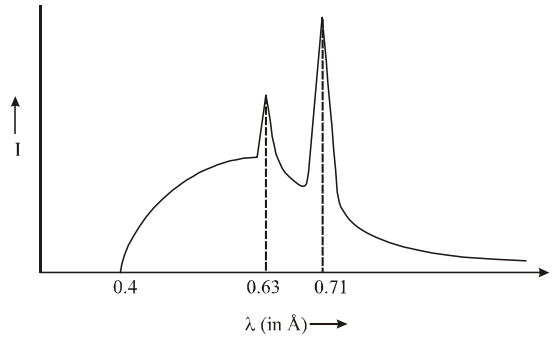
(A) $\frac{5}{9}mv_0^2$ (B) $\frac{4}{9}mv_0^2$

(C) $\frac{2}{9}mv_0^2$ (D) $\frac{mv_0^2}{9}$

Space for Rough work

Paragraph for Questions 14 & 16

Figure shows intensity versus wavelength graph of X-rays coming from Coolidge-tube with molybdenum as anticathode:



The two peaks shown in graph correspond to K_α & K_β *X*-rays

14. Wavelength of L_α X-rays from Coolidge tube will be (approximately)
(A) 5.60Å (B) 4.26 Å
(C) 0.33 Å (D) 1.34 Å

15. Voltage applied across Coolidge tube in (approximately)
(A) 20 *kV* (B) 16 *kV*
(C) 31 *kV* (D) 18 *kV*

16. In the potential difference across the Coolidge tube is increased such that the cut off wavelength of x-ray becomes 0.2 Å. Find the value of $\lambda_\beta - \lambda_\alpha$ in this condition.
(A) 0.02 Å (B) 0.04 Å
(C) 0.06 Å (D) 0.08 Å

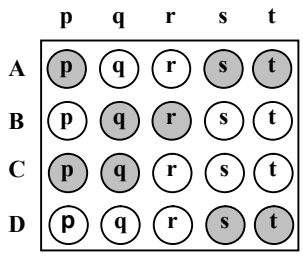
Space for Rough work

SECTION - B

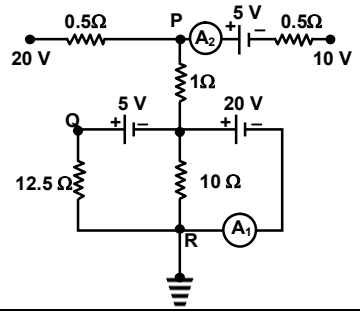
Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. In the adjacent circuit shown, all ammeters and batteries are ideal. Internal resistances of all the batteries are negligible. Point R is grounded. Then, match the following.



Column – I		Column – II	
(A)	Potential of point P	(p)	6 SI unit
(B)	Potential of point Q	(q)	25 SI unit
(C)	Reading of ammeter A_1	(r)	18 SI unit
(D)	Reading of ammeter A_2	(s)	47 SI unit

Space for Rough work

2. A lens is made up of a material of refractive index μ which is completely immersed in a medium of refractive index μ_0. The power of lens is P. Match the properties in column I with the corresponding type(s) of lens in column II

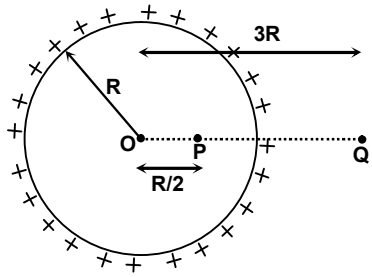
Column I		Column II	
(A)	$P \geq 0$ and $\mu > \mu_0$	(p)	Bi-convex
(B)	$P \geq 0$ and $\mu < \mu_0$	(q)	Bi-concave
(C)	$P \leq 0$ and $\mu > \mu_0$	(r)	Convexo concave
(D)	$P \leq 0$ and $\mu < \mu_0$	(s)	Concavo convex
		(t)	Plano convex

SECTION – C

(Integer Answer Type)

This section contains **6 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. Conducting shell of radius R carrying charge Q, another charge Q is placed at 3R distance from centre of shell. If the potential at point P due to charge on shell is $\frac{11}{3n}\frac{KQ}{R}$, Then the value of n is

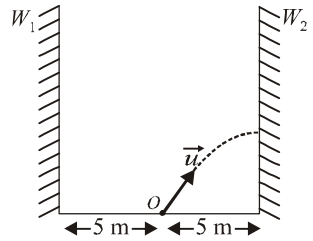


2. A liquid is kept in a cylindrical vessel. When the vessel is rotated about its axis, the liquid rises at its sides. If the radius of the vessel is 0.05 m and the speed of rotation is 2 revolutions per second, the difference in the heights (in cm) of the liquid at the centre and at the sides of the vessel is (take g = 10 ms^{-2} and $\pi^2 = 10$)

Space for Rough work

3. When a resistance R is connected in series with an element A, the electric current is found to be lagging behind the voltage by V angle 30° When the same resistance is connected in series with element B, current leads by 60°. When R, A, B are connected in series, the current new leads voltage by θ. Which is equal to $\tan^{-1}(K/\sqrt{3})$, then the value of K is (assume same AC source is used in all cases)

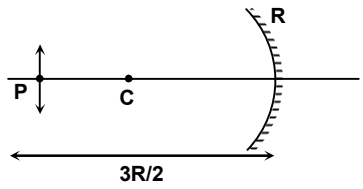
4. A ball of mass *m* = 2kg thrown from ground form point '*O*' between two walls, W_1 & W_2 at a distance 5 cm from '*O*' an shown in figure. Had the wall w_2 not been there range of ball would have been 20 m. Horizontal component of velocity of ball at the time of projection was 5 m/s. All collisions are elastic. Average force acting on one of the walls (in newton) is



5. A small object at P oscillates infront of a concave mirror as shown in the figure perpendicular to principal axis according to equation.

$y = 4 \sin\left(\omega t + \frac{\pi}{3}\right)$ cm,

y is the displacement of particle perpendicular to principal axis. The amplitude (in cm) of image is



6. A particle moves in a circle with a uniform speed, when it goes from a point A to a diametrically opposite point B, the momentum of the particle changes by $\vec{P}_A - \vec{P}_B = 2\text{kgm/s}(\hat{j})$ and the centripetal force acting on it changes by $\vec{F}_A - \vec{F}_B = 8\text{N}(\hat{i})$ where $\hat{i}, \hat{j}$ are unit vectors. Then the angular velocity (in rad/s) of the particle is

Space for Rough work

Chemistry

PART – II

SECTION – A
Straight Objective Type

This section contains **7 multiple choice** questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONLY ONE** option is correct.

1. At what $\frac{[Br^-]}{\sqrt{[CO_3^{2-}]}}$ does the following cell have its reaction at equilibrium?

Ag(s) $|Ag_2CO_3(s)|$ $Na_2CO_3(aq)|$ |KBr(aq)| AgBr(s)| Ag(s)

$K_{sp} = 8 \times 10^{-12}$ for Ag_2CO_3 and $K_{sp} = 4 \times 10^{-13}$ for AgBr

(A) $\sqrt{1}\times10^{-7}$ (B) $\sqrt{2}\times10^{-7}$

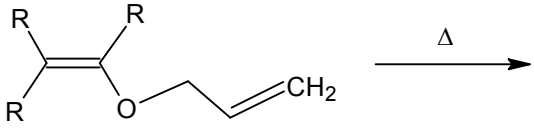
(C) $\sqrt{3}\times10^{-7}$ (D) $\sqrt{4}\times10^{-7}$

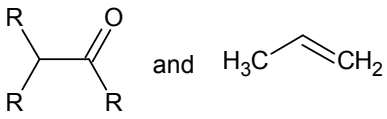
2. A human body excretes [removes by waste discharges, sweating etc] certain material by a law similar to radio activity. If technitium is injected in some form in the human body, the body excrets half of the amount in 24 hours. A patient is given an injection confainingTc^{99}. This isotope is radioactive with a half life 6 hrs. The activity from the body just after the injection is 6 μC. How many hours will elapse before the activity falls to 3 μCi?

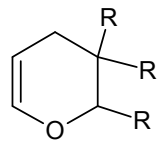
(A) 24 hours (B) 6 hours

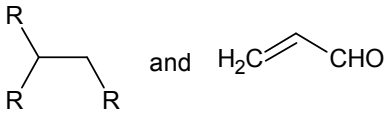
(C) 4.8 hours (D) 12 hours

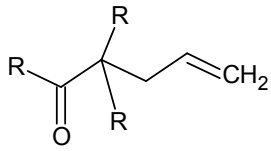
3. Find the product of the following reaction



(A) 

(B) 

(C) 

(D) 

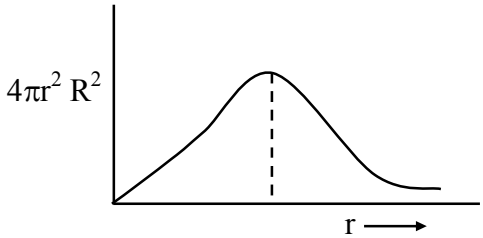
4. A big RED spherical balloon (radius = 6a) is filled up with gas. On this balloon six small GREEN spherical balloons (radius = a) are stuck on the surface in a specific manner. As RED balloon is slowly deflated, a point comes when all these six GREEN balloons touch and green balloons arrange themselves in a 3-D closed packing arrangement. At that stage the radius of the RED balloon would have reduced by approximately.

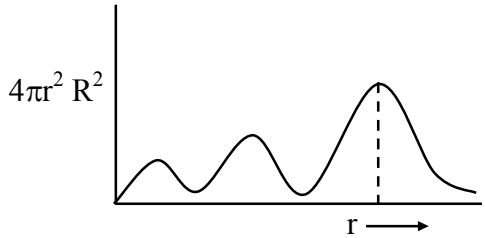
(A) 14.5 times (B) 1.414 times

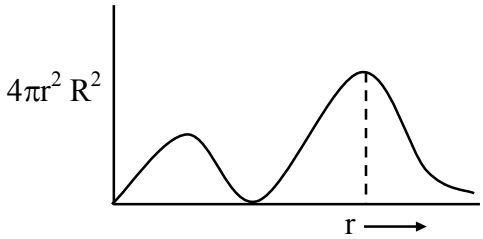
(C) 6.0 times (D) 2.42 times

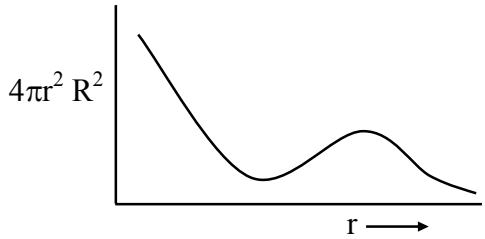
Rough Work

5. Which of the following graphs represents the radial charge density for the electron of outer most sub shell of Cu^{+2}?

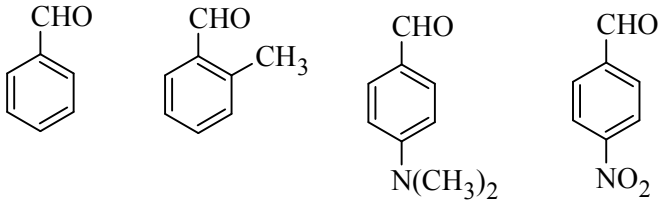
(A) 

(B) 

(C) 

(D) 

6. The order of reactivity towards Perkin reaction of the following aromatic aldehydes is



(a) (b) (c) (d)

(A) (d) > (a) > (b) > (c)
(B) (c) > (a) > (b) > (d)
(C) (b) > (c) > (a) > (d)
(D) (d) > (a) > (c) > (b)

7. Which of the following represents the correct order for the wavelength of absorption in the visible region for the following

$\underset{(I)}{[Ni(NO_2)_6]^{4-}}$ $\underset{(II)}{[NiF_6]^{4-}}$ $\underset{(III)}{[Ni(H_2O)_6]^{2+}}$ $\underset{(IV)}{[Ni(NH_3)_6]^{2+}}$

(A) I > IV > III > II
(B) II > IV > III > I
(C) II > III > IV > I
(D) I > III > II > IV

Rough Work

Multiple Correct Choice Type

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

8. A + B $\longrightarrow$ C + D is a stoichiometrically balanced reaction. The initial rate of the reaction is doubled if the initial concentration of A is doubled, but is quadrupled if the initial concentration of B is doubled.
Select the correct statement(s)
(A) The reaction is first order in B and second order in A
(B) The reaction is first order in A and second order in B
(C) The reaction can not be a single-step reaction
(D) The overall order of the reaction is 3

9. In which of the following cases will a violet colouration be observed
(A) An alkaline solution of sodium nitroprusside is treated with a solution of Na_2S
(B) A solution of sodium cobaltinitrite is treated with KCl
(C) A solution of $Mn(NO_3)_2$ is treated with sodium bismuthate or red lead in the presence of concentrated HNO_3
(D) A solution of sodium nitroprusside in aqueous, NaOH is heated ith Na_2SO_3

10. $CH_3-CH=NH \xrightarrow{?} CH_3CH_2-NH_2$ for this conversion suitable reagent
(A) $Na-Hg/H_2O$ (B) H_2/Pd
(C) $LiAlH_4/Et_2O$ (D) $NaBH_4/EtOH$

11. $X + H_2SO_4 \rightarrow Y$ (a colourless gas with pungent smell)
$Y + K_2Cr_2O_7 + H_2SO_4 \rightarrow$ Green solution
(A) X is SO_3^{2-} (B) X is CO_3^{-2}
(C) Y is H_2S (D) Y is SO_2

Rough Work

Comprehension Type

This section contains **2 paragraphs.** Based upon the first paragraph 2 multiple choice questions and based upon the second paragraph **3 multiple choice questions** have to be answered. Each of these questions has four choices A), B), C) and D) out of **WHICH ONLY ONE IS CORRECT.**

Paragraph for Question Nos. 12 to 13

The process of eudiometry is used to calculate the volume of the various reacting gases present together in a closed vessel, it is done by sparking the content and noting the volume change or pressure change depending upon the conditions mentioned. Various reagents like alkaline pyrogallol, turpentine oil, caustic potash, conc. H_2SO_4 are used to absorb the selective gases like O_2, O_3, CO_2 and H_2O (v) respectively.
A 2.0 mole mixture of $H_2(g)$, $O_2(g)$ and He(g) are placed together in a closed container at pressure equal to 50 atm. An electric spark is passed and pressure is noted as 12.5 atm after cooling the content to original temp. (room temp). Oxygen gas is introduced for pressure to change to 25 atm, keeping volume and temp constant. Again electrical spark is passed and pressure drops to 10 atm under original temp. conditions.

12. The moles of He(g) present in the mixture sample is
(A) 0.2 (B) 0.1
(C) 0.5 (D) 1.4

13. The mole fraction of $H_2(g)$ present in the initial mixture is:
(A) 0.05 (B) 0.7
(C) 0.5 (D) 1.25

Rough Work

Paragraph for Question Nos. 14 to 16

An unknown organic compound (A) with the molecular formula $C_9H_{12}O$ does not decolourize Br_2 in CCl_4, reacts with sodium to give a colourless, odourless gas and can be oxidised by hot alkaline $KMnO_4$ to give benzoic acid. Then, following test were performed on the compound (A) is given sequence:

Test (a): The colour of the solution changes from orange to green when $K_2Cr_2O_7$ is added to it.

Test (b): The compound can be resolved.

Test (c): No yellow precipitate of CHI_3 is formed when reacted with I_2 and OH^-.

Test (d): Oxidation of the compound with CrO_3 in pyridine gives another compound that gives Tollen's test.

14. Which of the following would be the structure of compound if test (a) is negative?
(A) $PhCH(OH)CH_2CH_3$ (B) $PhC(OH)(CH_3)_2$
(C) $PhCH_2CH(OH)CH_3$ (D) $PhCH(CH_3)CH_2OH$

15. Which of the following would not be the structure of compound (A) after test (b)?
(A) $PhCH(OH)CH_2CH_3$ (B) $PhCH_2CH(OH)CH_3$
(C) $PhCH_2CH_2CH_2OH$ (D) $PhCH(CH_3)CH_2OH$

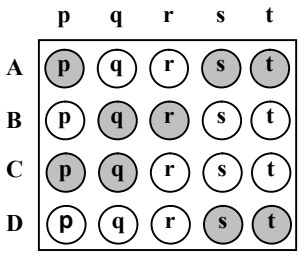
16. The structure assigned to compound (A) after the test (d) would be
(A) $PhCH(OH)CH_2CH_3$ (B) $PhCH_2CH(OH)CH_3$
(C) $PhCH(CH_3)CH_2OH$ (D) $PhC(OH)(CH_3)_2$

Rough Work

SECTION-B
(Matrix-Match Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

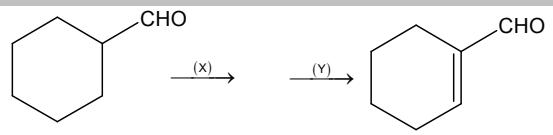
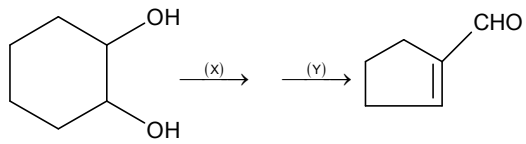
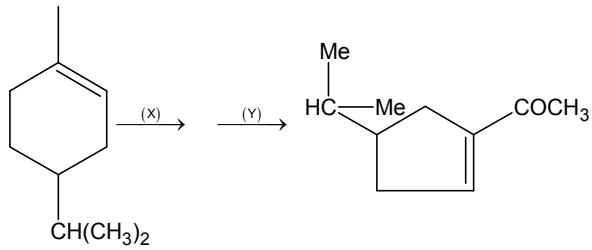
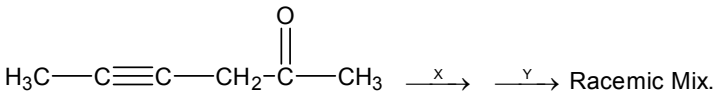


1. Match the following metals given in **Column I** with the appropriate metal extraction process(es) listed below in **Column II**.

Column – I	**Column – II**
(A) Silver	(p) Fused salt electrolysis
(B) Lead	(q) Cyanide process
(C) Iron	(r) Carbon monoxide reduction
(D) Magnesium	(s) Self reduction
	(t) Frothflotation

Rough Work

2. Column – I contain reaction and Column – II consist of reagent used in these reaction. Match the following.

Column – I	Column – II
(A) 	(p) X = Na/liq. NH_3 Y = cold dil. $KMnO_4/OH^-$
(B) 	(q) X = Br_2, $CHCl_3$ Y = CH_3ONa/CH_3OH
(C) 	(r) X = O_3/H_2O, Zn Y = NaOH/Δ
(D) 	(s) X = HIO_4 Y = NaOH/Δ
	(t) X= H_2/Pd Y=Zn-Hg/ conc. HCl

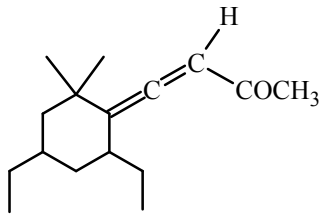
Rough Work

SECTION –C
Integer Answer Type

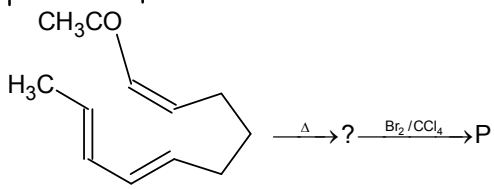
This section contains **6 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. How many of the following reactions are addition-elimination reactions of carbonyl compounds?
(I) Cyanohydrin formation
(II) Acetal formation
(III) Semicarbazone formation
(IV) Aldol formation
(V) Hemiacetal formation
(VI) Oximeformation
(VII) Ketal formation
(VIII) Imine formation
(IX) 2,4-DNPderivative formation

2. Total number of stereoisomers of the compound given below is x



And number of chiral carbons are there in the product 'P'is y



then the value of (x+y)/2 is?

3. A 50 ml 1.92% (W/V) solution of a metal ion M^{n+} (atomic weight = 60) was treated with 5.332 g hydrazinehydrate (90% pure) and mixture was saturated with CO_2 gas when entire metal gets precipitated as a complex $[M(H_2N - NHCOO)_n]$. The complex was filtered off and the filtrate was titrated with $\frac{M}{10}KIO_3$ in the presence of conc. HCl according to the following equation:

$$N_2H_4 + IO_3^- + 2H^+ + 2Cl^- \longrightarrow ICl + 3H_2O + N_2 \uparrow$$

The volume of $\frac{M}{10}KIO_3$ solution needed for the end point to arrive was 480 ml. Find the value of n.

Rough Work

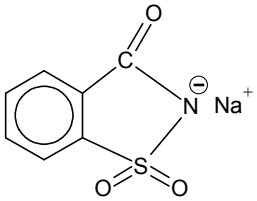
4. How many statements are true in the given question of about?
(i) orthoboric acid It is a strong tribasic acid.
(ii) orthoboric acid does not act as a proton donor but behaves like a Lewis acid by accepting OH^- ion.
(iii) orthoboric acid is prepared by the action of mineral acid on borax solution.
(iv) orthoboric acid has a layer structure in which hydrogen bonds unite $B(OH)_3$ units.

(v) If the number of revolutions made by electron in1.0 s in H atom in its nth orbit is twice of the number of revolution made by electron in 1.0 s in the 2nd orbit of H–atom, then n is 1.

(vi) Forms a colourless solution with coloured precipitate(r)$CrCl_3$(s) + dil HNO_3(aq)

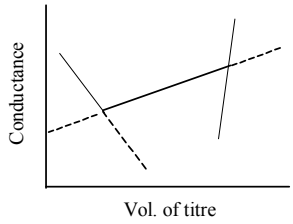
(vii) Forms a coloured solution with no precipitate(s)K_2CrO_4(aq) + $Ba(NO_3)_2$(aq)

(viii) F > N > C > Si >Ca - metallic character
(ix) Saccharin (pK_a = 2) is an artificial zero calorie sweetner added to toothpaste

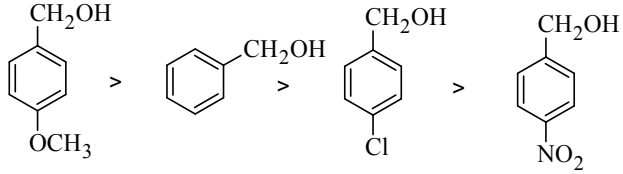


Sodium Saccharin Structure .Two flasks A&B both contain 18.3 gm of Saccharin dissolved in 100 ml solution. 2 grams of solid NaOH is added to A and 4 grams of solid NaOH is added to B. This resulted in the formation of the above structure in both flasks. The difference in pH of the two flasks after the experiment is: 6
(x) C > Si > P > N - electron affinity
(xi) O > N > F > C - second ionisation potential
(xii) $I_2 < F_2 < Br_2 < Cl_2$ Bond energy ORDER

(xiii) mixture of HCl and CH_3COOH titrated against NaOH the correct graphical representation is

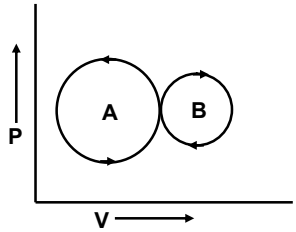


(xiv) The CORRECT ORDER OF basic character of the following alcohols is

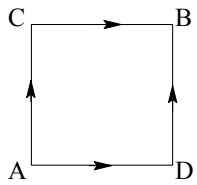


Rough Work

5. In the given graph, the area of circle A and B are 40 unit and 36 unit respectively, total work done 'X' unit.



When a system is taken from state A to B along path ACB, as shown in the figure, 5 unit of heat flows into the system and the system does 3 unit of work. If 'y'unit heat flows into the system along the path ADB, and the work done by the system is 2 unit ?



Then the value of x + y is

6. Alkaline hydrolysis of chloral hydrate produces a compound (A) which is used as a solvent and preservative for anatomical specimens and also has anesthetic properties. (A) is prone to aerial oxidation in presence of sunlight to give a poisonous compound (B) which may be made non poisonous by adding dilute solution of ethanol which converts it into a non poisonous compound (C). How many moles of CH_3MgBr will be added to (C) followed by acidic hydrolysis to give alcohol?

Rough Work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains 7 multiple choice questions numbered 1 to 7. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. The number of 6-digit numbers whose digits are selected from set {5, 6, 7, 8, 9} such that any digit that appears in the number appears at least twice is
   (A) 1100 (B) 1200
   (C) 1400 (D) 1405

2. The equation of common tangent to the curve $|z - 2i| = 2$ and $|z - 1 - 3i| + |z + 1 - 3i| = 2\sqrt{2}$ is
   (A) $z + \bar{z} = 4$ (B) $z - \bar{z} = 8i$
   (C) $z + \bar{z} = 8$ (D) $z - \bar{z} = 4i$

3. Let $f(n) = \sum_{r=0}^{n}\sum_{k=r}^{n} {}^{k}C_{r}$ then total number of divisors of f(9) is equal to
   (A) 5 (B) 6
   (C) 7 (D) 8

4. Let $2\left(1+x^3\right)^{100} = \sum_{i=0}^{100}\left\{a_i x^i - \cos\left(\frac{\pi}{2}(x+i)\right)\right\}$. If $\sum_{i=0}^{50} a_{2i} = 2^k$ then the value of k is
   (A) k = 100 (B) k = 200
   (C) k = 310 (D) k = 410

5. Let M = {(x, y) : $y \geq x^2$} and N = {(x, y) : $x^2 + (y - a)^2 \leq 1$}. The necessary and sufficient condition for $M \cap N = N$ is
   (A) $a \geq \frac{5}{4}$ (B) $a = \frac{5}{4}$
   (C) $a \in R^+$ (D) $a \in (0, 1)$

Space for rough work

6. The value of the ratio $\dfrac{\int_0^{\pi} x^3 \log_e(\sin x)\,dx}{\int_0^{\pi} x^2 \log_e\left(\sqrt{2}\sin x\right)dx}$ is

(A) $\dfrac{\pi}{2}$ (B) $\dfrac{2\pi}{3}$

(C) $\dfrac{3\pi}{4}$ (D) $\dfrac{3\pi}{2}$

7. A parabola $y = ax^2 + bx + c$ crosses the x-axis at $(\alpha, 0)$ and $(\beta, 0)$ both to the right of the origin. A circle also passes through these two points the length of tangent from the origin to the circle is

(A) $\sqrt{\dfrac{a}{b}}$ (B) $\sqrt{\dfrac{c}{a}}$

(C) $\sqrt{bc}$ (D) $\dfrac{b^2}{c^2}$

Multiple Correct Choice Type

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

8. Positive integers are written on all the faces of a cube, one on each. At each corner (vertex) of the cube, the product of the numbers on the faces that meet at the corner is written. The sum of the numbers written at all the corner is 2004. If T denotes the sum of the number on all the faces, the possible value of T are

(A) 174 (B) 175

(C) 339 (D) 505

9. If the equation $x^2 + (a - 2)x + 1 = 3|x|$ has exactly three distinct real solutions in x, the value of a is/are

(A) 1 (B) 2

(C) 3 (D) 4

Space for rough work

10. In an acute triangle ABC, O is circumcentre, H is orthocentre and G is the centroid. Let OD be perpendicular to BC and HE be perpendicular to CA, with D on BC, E on CA. Let F be the mid-point of AB. Suppose the area of triangle ODC, HEA and GFB are equal the possible value of angle C is

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{6}$ (D) $\frac{\pi}{2}$

11. A player throws an ordinary dice, whenever he throws 1, he gets an additional throw. Then probability to get sum of score n is

(A) $\frac{1}{5}\left(\frac{6^4-1}{6^4}\right)$ is n = 5 (B) $\frac{1}{5}\left(\frac{6^5-1}{6^5}\right)$ if n = 6

(C) $\frac{1}{30}\left(\frac{6^5-1}{6^5}\right)$ if n = 7 (D) 0 if n = 1

Comprehension Type

This section contains 2 paragraphs. Based upon one of paragraphs 2 multiple choice questions and based on the other paragraph 3 multiple choice questions have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Paragraph for Question Nos. 12 to 13

Read the following write up carefully and answer the following questions:
A trapezium ABCD, in which AB is parallel to CD, is inscribed in a circle with centre O. Suppose the diagonal AC and BD of trapezium intersect at M and OM = 2

12. If $\angle AMB = 60^\circ$ then difference between length of parallel sides is

(A) $\sqrt{3}$ (B) 2

(C) $2\sqrt{3}$ (D) 4

13. If $\angle AMD$ is 60° then difference between length of parallel sides is

(A) 2 (B) $\sqrt{3}$

(C) $2\sqrt{3}$ (D) $4\sqrt{3}$

Space for rough work

Paragraph for Question Nos. 14 to 16

Read the following write up carefully and answer the following questions:

An ellipse whose major axis is parallel to x-axis such that segment of a focal chord are 1 and 2 units. The line $\frac{x}{a}+\frac{y}{b}+\frac{1}{c}=0$ are the chords of the ellipse such that a, c, b are in H.P. and are bisected by the point at which they are concurrent, the equation of auxiliary circle is $2x^2 + 2y^2 + 4px + 4qy - 16p + 1 = 0$ then

14. The centre of ellipse is

(A) (1, –1) (B) (1, 1)

(C) $\left(\frac{1}{2},\frac{1}{2}\right)$ (D) $\left(-\frac{1}{2},-\frac{1}{2}\right)$

15. Equation of director circle is

(A) $x^2 + y^2 - x - y - \frac{37}{6} = 0$ (B) $x^2 + y^2 + x + y + \frac{37}{6} = 0$

(C) $x^2 + y^2 + x + y - \frac{37}{6} = 0$ (D) $x^2 + b^2 - x - y + \frac{37}{6} = 0$

16. Eccentricity of ellipse is

(A) $\frac{1}{3}$ (B) $\frac{1}{\sqrt{3}}$

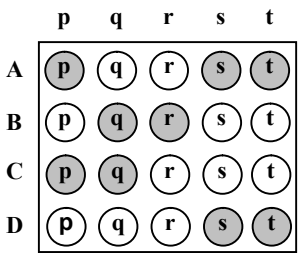
(C) $\sqrt{\frac{2}{3}}$ (D) $\frac{2}{3}$

Space for rough work

SECTION-B
(Matrix-Match Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. Match the following Column-I with Column-II

	Column – I		Column – II
(A)	The number of all triples (a, b, c) such that all three term a, b, c are in harmonic progression in which a = 20 and b divides c (where a, b, c are strictly increasing positive integers)	(p)	66
(B)	The number of solutions of m + n + p = 10, (m, n, p are non-negative integers)	(q)	20
(C)	The number of ordered pair (a, b) satisfying the equation ab + b – a + 1 = 0 are where a, b $\in$ I	(r)	4
(D)	The number of ordered pairs (x, y) of real numbers such that $(x + iy)^{20} = x - iy$ is (where $i = \sqrt{-1}$)	(s)	5
		(t)	22

2. Match the following Column-I with Column-II

	Column – I		Column – II
(A)	Let $f(x) = x^4 + ax^3 + bx^2 + cx + d$ (where a, b, c, d are real coefficient) and f(x) = 0 has real roots, If |f(i)| = 1 and a = b = c = d = k then k can be (where $i = \sqrt{-1}$)	(p)	0
(B)	If $x^5 = 1$ ($x \neq 1$) then $\frac{x}{1+x^2} + \frac{x^2}{1+x^4} + \frac{x^3}{1+x} + \frac{x^4}{1+x^3}$ equals	(q)	1
(C)	Let $f(x) = \ln\left[\cos\lvert x\rvert + \frac{1}{2}\right]$ (where [.] denotes the greatest integer function), then $\int_{x_1}^{x_2}\left(\lim_{n\to\infty}\left(\frac{(f(x))^n}{x^2+\tan^2 x}\right)\right)dx$ is (where $x_1, x_2 \in \left[-\frac{\pi}{6}, \frac{\pi}{6}\right]$)	(r)	2
(D)	Let ABCD be a cyclic quadrilateral inscribed in a circle of radius 2, such that BD = $2\sqrt{3}$, AB = 1. If $\angle$BAD is acute and AD = $a + b\sqrt{5}$, then the value of a + b is equal to	(s)	3
		(t)	4

Space for rough work

SECTION – C

(Integer Answer Type)

This section contains **6 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. Let X be the set of all positive integers greater then or equal to 8 and let f : X $\to$ X be a function such that f(x + y) = f(xy) for all $x \geq 4$, $y \geq 4$. If f(8) = 9 then f(9) is equal to _____

2. If f(x) and g(x) be two function such that $f''(x) = 2f'(x) - 2f(x)$, $g(x) = f'(x) - f(x)$. Let F(x) be defined as $F(x) = [f^2(x) + g^2(x)][f^2(-x) + g^2(-x)]$. If F(0) = 1 then the value of F(1) is _____

3. Let $f(x) = \tan^{-1}(\cot x - 2\cot 2x)$ and $\sum_{r=1}^{5} f(r) = a - b\pi$ where a, b $\in$ N, then the value of b is _____

4. The least possible area of convex quadrilateral having two vertices on two branches of hyperbola xy = 2 and other two vertices on the two branches of xy = –2 is _____

5. Consider two polynomial f(x) and g(x) given by $f(x) = \sum_{r=0}^{200} \beta_r x^r$ and $g(x) = \sum_{r=0}^{200} \alpha_r x^r$ such that $\beta_r = 1$ $\forall$ $100 \leq r \leq 200$ and f(x + 1) = g(x). Let $A = \sum_{r=100}^{200} \alpha_r$, then the remainder when A is divided by 15 is equal to _____

6. Consider a square ABCD of diagonal length $2\sqrt{3}$. The square is folded along the diagonal AC so that the plane of ΔABC is perpendicular to the plane of ΔADC. The shortest distance between AB and CD is _____

Space for rough work